Testing of ShopClues E-Commerce Website

A PROJECT(25CAP-602) REPORT SUBMITTED IN PARTIAL

Submitted by

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CERTIFICATE

This is to certify that Sarvesh Yadav has successfully completed the project titled "Software Testing of Web-Based Application" as a part of the academic requirements for the degree of Master of Computer Applications (MCA).

This project report has been submitted in partial fulfillment of the requirements for the award of the degree under the guidance of the faculty. The work presented in this project is the original work of the student and has not been submitted elsewhere for any other degree or diploma.

The project demonstrates the application of software testing concepts, including test case design, execution, and analysis of results. The student has shown sincere effort and dedication in completing this project.
We wish him success in all his future endeavors.

Project Guide

Mrs. Priyanka

Assistant professor

University Institute of Computing

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout the development of this project.
First and foremost, I would like to thank my respected teachers and mentors for their valuable guidance, encouragement, and continuous support. Their knowledge and suggestions helped me understand the concepts clearly and complete this project successfully.

I am also thankful to my institution for providing me with the necessary resources and environment to work on this project. The facilities and learning atmosphere played an important role in enhancing my skills and knowledge.
I would like to extend my heartfelt thanks to my friends and classmates for their cooperation, ideas, and motivation during the project work. Their support made the process easier and more enjoyable.

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INTRODUCTION

Software testing is a crucial phase in the Software Development Life Cycle (SDLC) that ensures the quality, reliability, and performance of an application. The primary objective of testing is to identify defects, verify functionality, and ensure that the system meets the specified requirements before deployment. A well-tested system improves user satisfaction, reduces maintenance costs, and enhances overall system efficiency.
This project focuses on testing a web-based application that includes key modules such as Login, Signup, Product Search, Cart Management, and Checkout. These modules represent the core functionalities of an e-commerce system and are essential for providing a smooth and secure user experience. The testing process involves validating user inputs, checking system responses, and ensuring proper handling of both valid and invalid scenarios.

Different types of testing techniques are applied in this project, including functional testing, validation testing, and basic security testing. Functional testing ensures that each feature works according to the requirements. Validation testing checks input fields such as email, password, and mobile number to ensure proper data handling. Security testing is performed to prevent issues such as SQL injection and unauthorized access.

TESTING METHODOLOGY

- The testing methodology follows a structured approach to ensure that the application meets all functional and non-functional requirements.
- The process begins with understanding the system requirements and identifying key modules such as Login, Signup, Product Search, Cart Management, and Checkout.
- Test cases are designed for each module to cover both **positive (valid input)** and **negative (invalid input)** scenarios.
- Each test case includes details such as **Test Case ID, Description, Input, Expected Output, Actual Result, and Status (Pass/Fail)**.

a)TEST PLAN

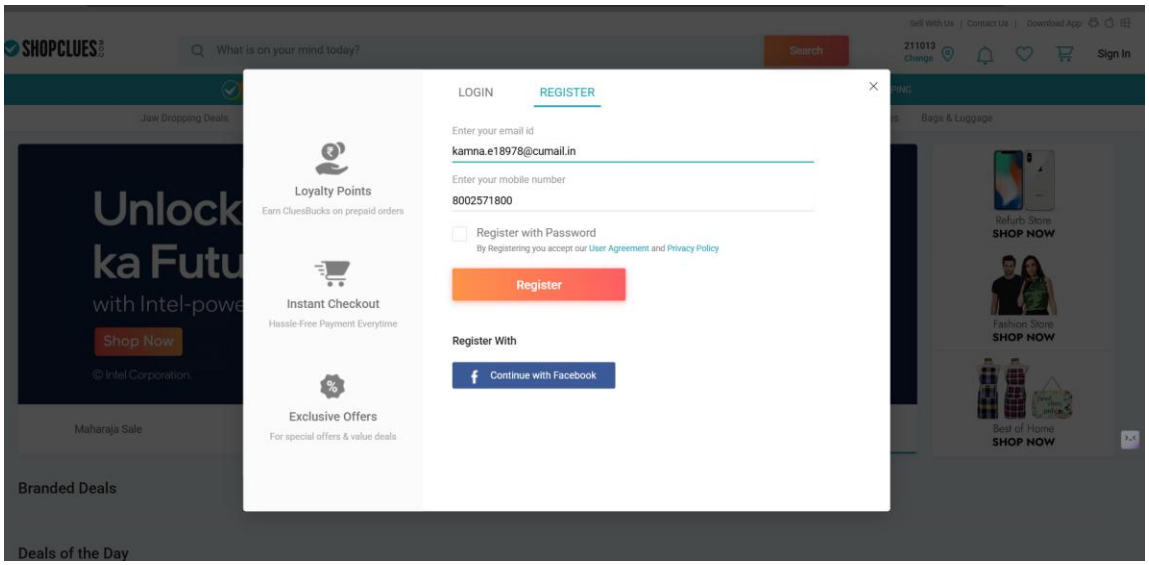
1)REGISTER TEST CASES

TC ID	Description	Input	Expected Output	Actual Result	Result
TC_S01	Valid signup	sarveshyadav0046@gmail.com	Success	Signup successful	Pass
TC_S02	Unknown email	Sarvesh46@gamil.com	Already have account	Already registered	Pass
TC_S03	Invalid email	sarvesh@	Error	Error shown	Pass
TC_S04	Invalid Mo.no	12345	Invalid	Invalid number	Pass
TC_S05	OTP verify	4578	Verified	OTP verified	Pass
TC_S06	Strong password	Sarvesh*1	Success	Accepted	Pass
TC_S07	Weak password	111111	Invalid weak password	Accepted	Fail
TC_S08	Email not registered	Sarvesh1212@gmail.com	Invalid	Invalid message	Pass

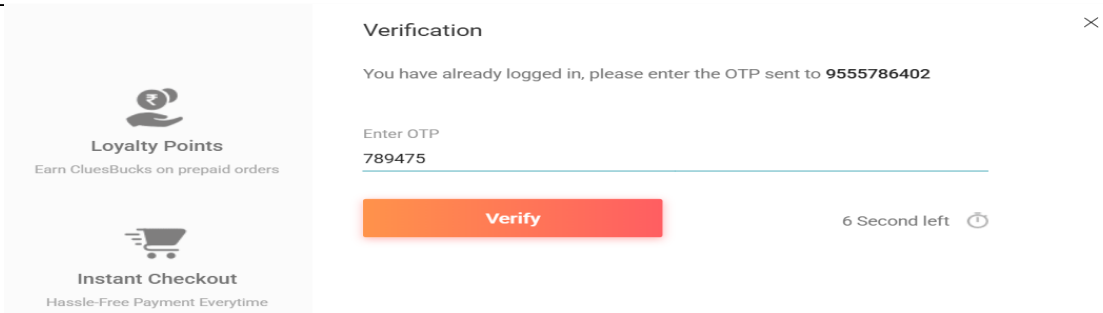
a)Decision Table

Conditions / Rules	**R1**	**R2**	**R3**	**R4**	**R5**	**R6**	**R7**	**R8**
Valid Email	Y	N	N	Y	Y	Y	Y	N
Email Exists	N	Y	-	-	-	-	-	N
Valid Mobile No.	Y	Y	Y	N	Y	Y	Y	Y
OTP Verified	Y	Y	Y	Y	Y	Y	Y	Y
Strong Password	Y	Y	Y	Y	Y	Y	N	Y
Email Registered	Y	Y	Y	Y	Y	Y	Y	N
Action	Signup Success	Reject	Error	Reject	Verify	Accept	Reject	Reject

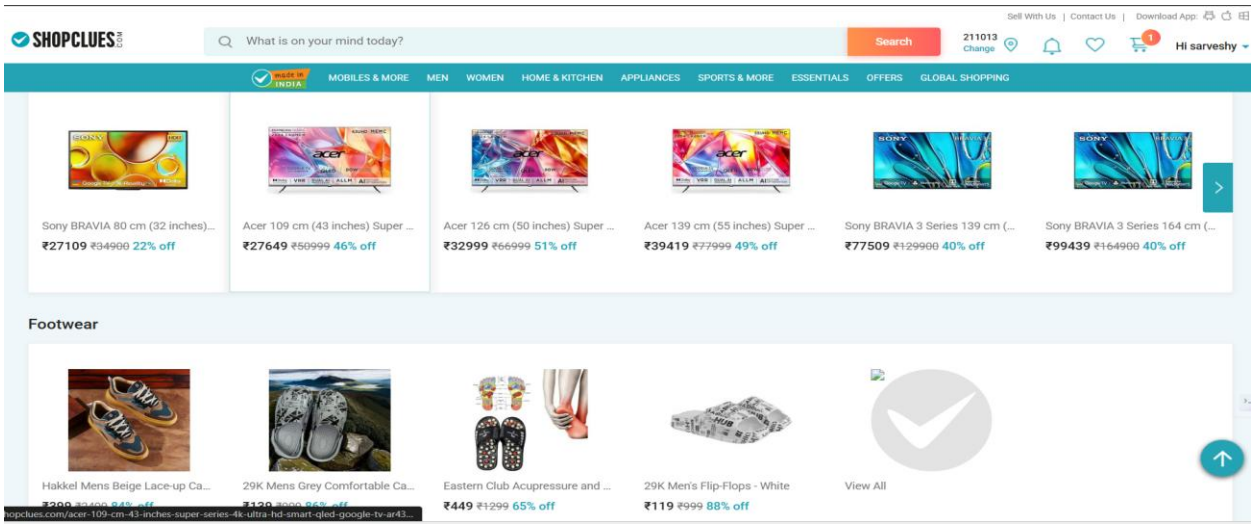
i) User Registration Interface – Entering Email and Mobile Number for Account Creation with Loyalty Benefits and Social Login Options



ii) OTP Verification Screen – User Entering One-Time Password with Countdown Timer for Secure Login Authentication



iii)



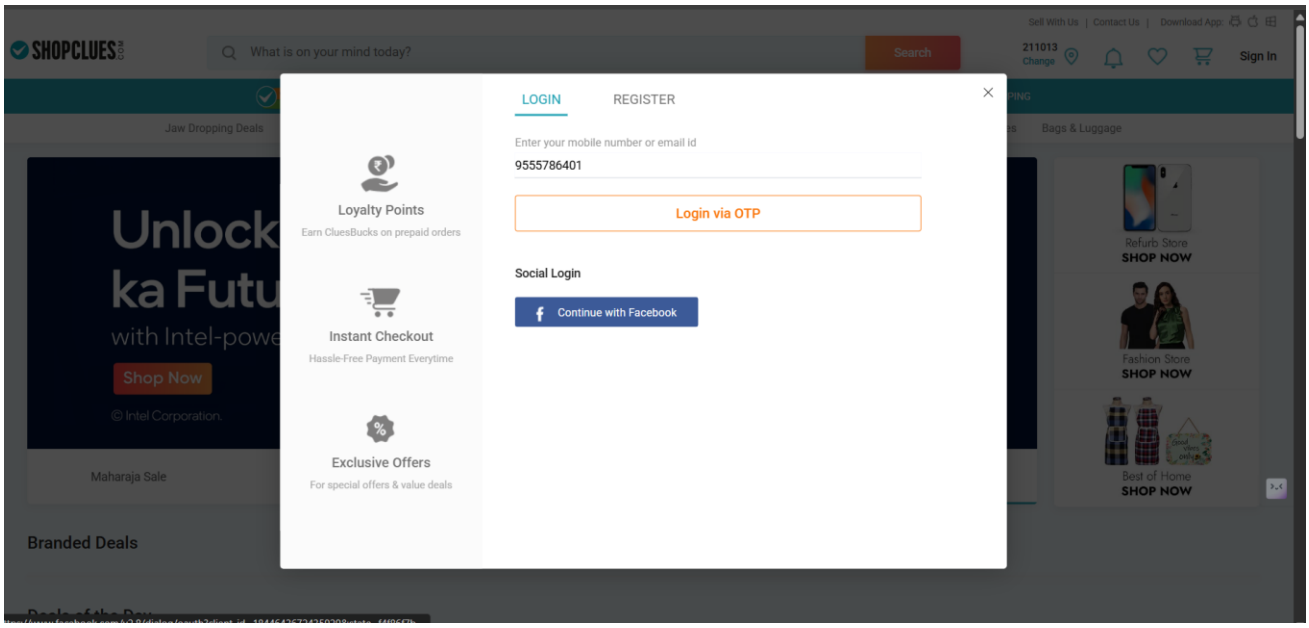
2)Login page

TC ID	Description	Input	Expected Output	Actual Result	Result
TC_L01	Valid email	Sarveshyayadav46@gmail.com	Success	Login successful	Pass
TC_L02	Invalid password	564	Error	Error shown	Pass
TC_L03	Empty fields	Blank	Validation	Validation shown	Pass
TC_L04	Invalid email	sarves@	Fail login	Login blocked	Pass
TC_L05	Mo. No. (valid)	9555786401	pass	pass	Pass

a)Decision Table

Conditions / Rules	**R1**	**R2**	**R3**	**R4**	**R5**
Valid Email	Y	Y	N	N	Y
Correct Password	Y	N	-	-	Y
Fields Filled	Y	Y	N	Y	Y
Valid Mobile No.	-	-	-	-	Y
Action	Succes	Error	Validation	Block	Login Success

i) <u>Login page</u>



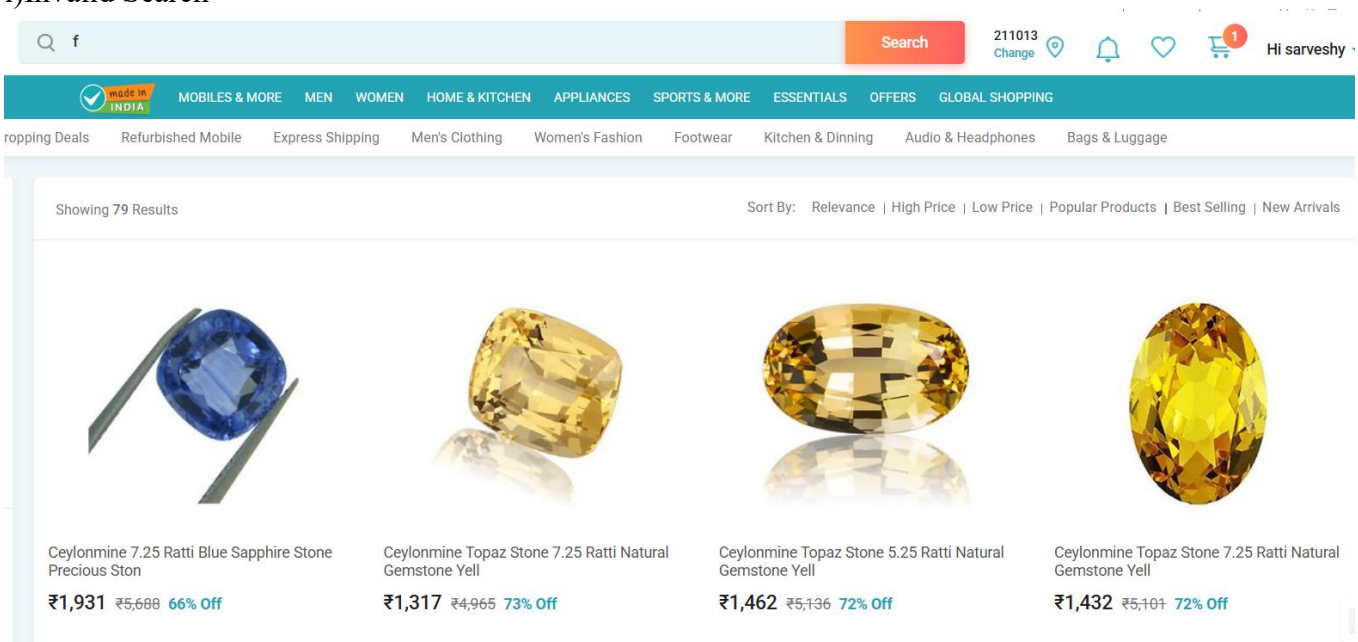
3)SEARCH & CART TEST CASES

TC ID	Description	Input	Expected Output	Actual Result	Result
TC_P01	Valid search	'Shoes'	Results	Results shown	Pass
TC_P02	Invalid search	'F'	No results	results	Fail
TC_P03	Operator (search)	'((!'	invalid	Fail	Pass
TC_PO4	Blank Search	""	No result	No result	Pass
TC ID	**Description**	**Input**	**Expected Output**	**Actual Result**	**Result**
TC_C01	Add item	Click	Added	Item added	Pass
TC_C02	Add without login	Guest	Prompt	Login prompt	Pass
TC_C03	Out of stock	Select	Error	Error shown	Pass

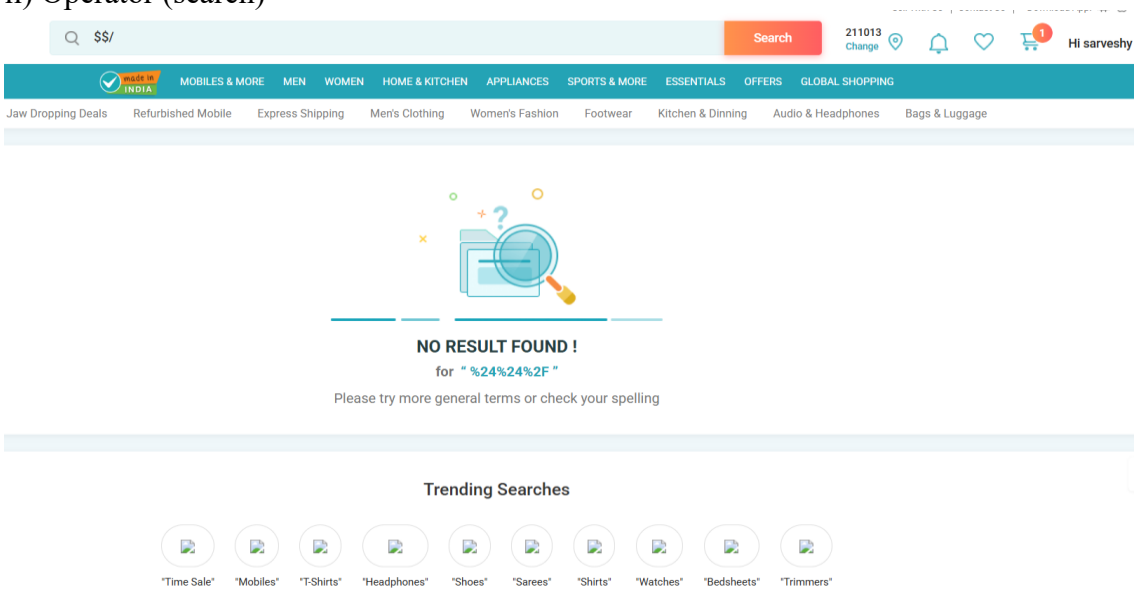
a) Decision Table

Conditions / Rules	**R1**	**R2**	**R3**	**R4**
Valid Input	Y	N	N	N
Special Characters	N	N	Y	N
Blank Input	N	N	N	Y
Action	Show Results	No Result	Error	No Result

i)Invalid Search



ii) Operator (search)

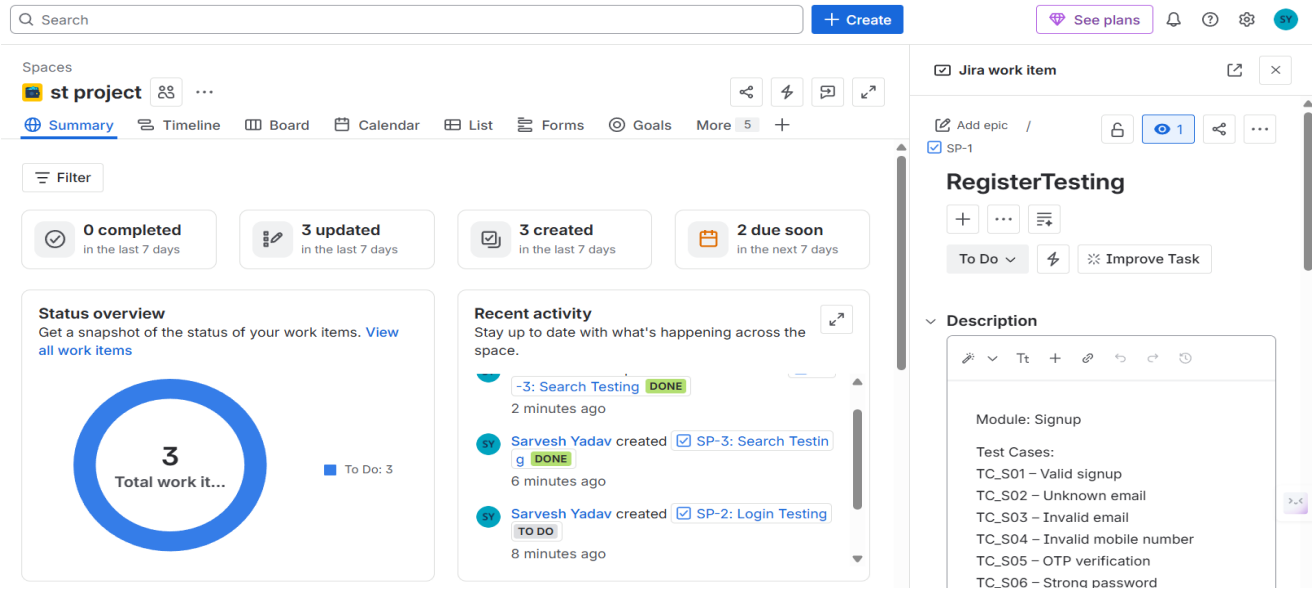


BUG REPORT

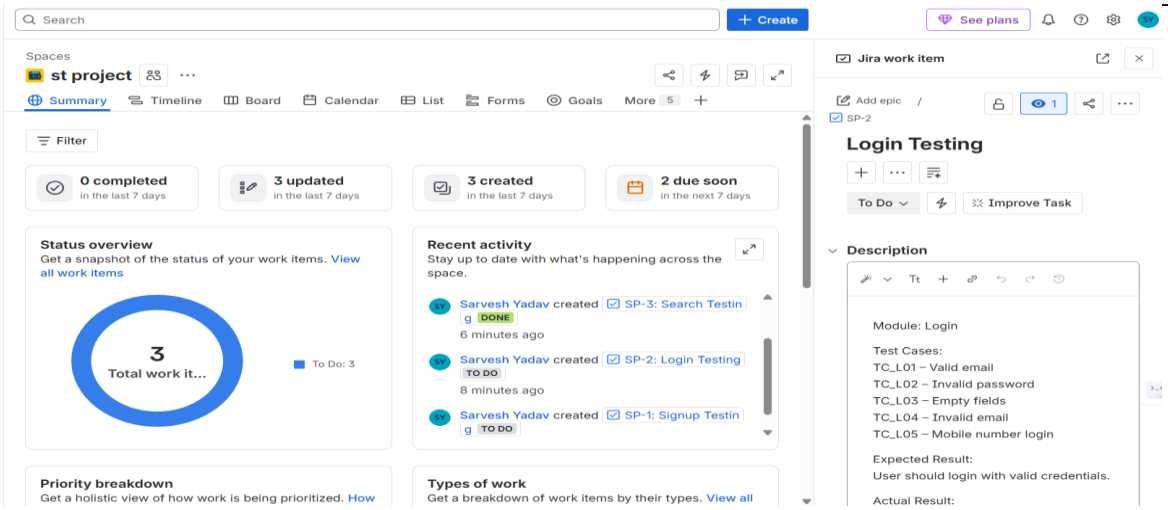
Bug ID	Description	Severity	Status
B01	Filter issue	Medium	Open
B02	Slow loading	High	Open
B03	Checkout bug	High	Open
B04	UI issue	Medium	Open

Creating a new task in JIRA

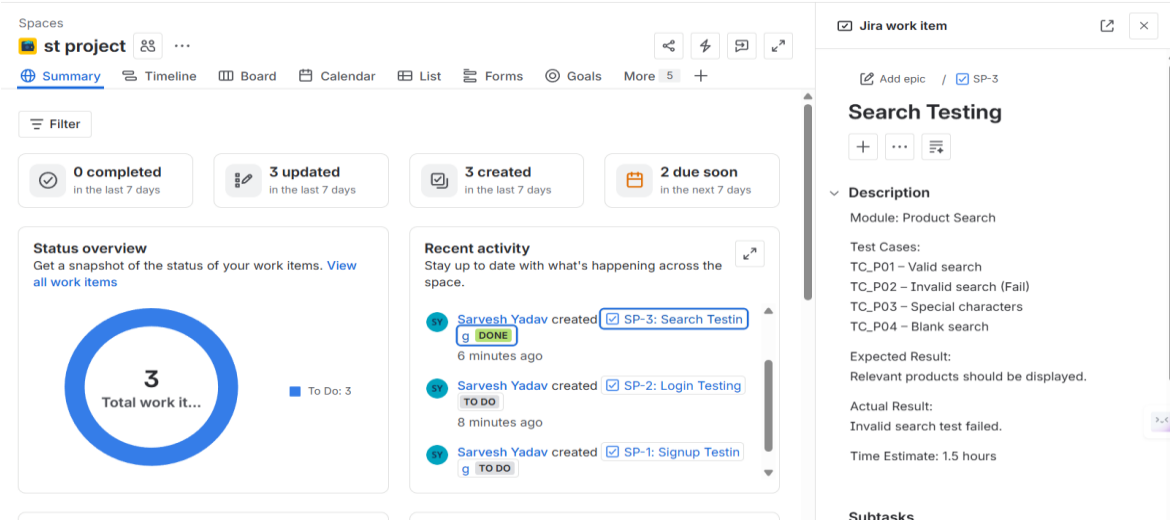
i) Jira Dashboard Displaying Registration Testing Task with Defined Test Cases, Status Tracking, and Progress Monitoring



ii) Jira Dashboard Displaying Login Testing Task with Defined Test Cases, Status Tracking, and Execution Details

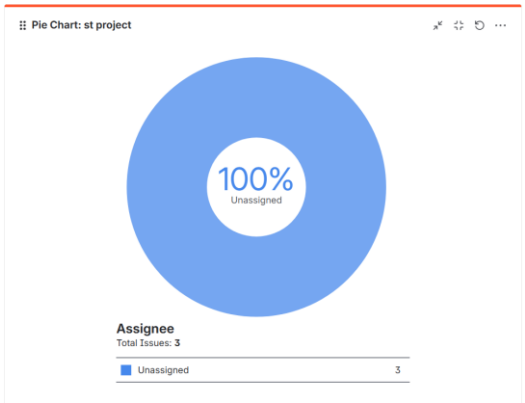


iii) Jira Dashboard Displaying Product Search Testing Task with Defined Test Cases, Status Tracking, and Execution Results



Gadgets-

i) Pie Chart



Conclusion

The software testing process carried out in this project successfully evaluated the functionality, performance, and reliability of the ShopClues e-commerce web application. All major modules, including Login, Signup, Product Search, Cart Management, Checkout, and User Interface, were thoroughly tested using well-structured test cases. This systematic approach ensured that both valid and invalid scenarios were covered effectively.

The results indicate that the majority of the core functionalities are working correctly and efficiently. Features such as user authentication, product search, cart operations, and order placement performed as expected under normal conditions. This demonstrates that the application is stable and capable of handling basic user requirements.

Git hub link-